Roll No. _____

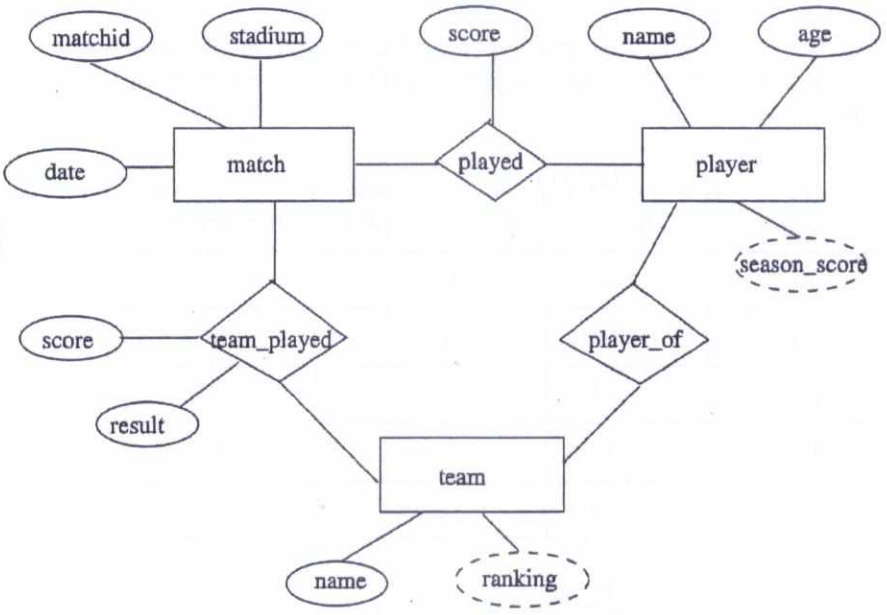
Thapar Institute of Engineering and Technology
Electronics and Communication Department

Course Name: Database Management Systems
Date & Day: 24th September 2025, Wednesday
Name of Faculty: Dr. Ashu Sharma, Dr. Deepak Rakesh, Dr. Amit Kumar

Course Code: UEC716/ UCS310
Time: 3:00 - 5:00 PM
Max Marks: 30

Note: Attempt all questions, Assume any missing value.

PLEASE MIND TO ATTEMPT ALL SUBPARTS OF A QUESTION AT ONE PLACE.

Q.No.	Questions	Marks	CL	BL
Q1(a)	Discuss the different types of DBMS Architecture.	5	C1	L1
Q1(b)	Consider the relation schema R (A, B, C, D, E, F) where FDs = {AB -> D, B-> C, AE -> C, A -> C, A -> D, D -> EF} Compute canonical cover and list candidate keys. Note: Show the steps.	5	C3	L3
Q2(a)	Consider the following E-R diagram and answer the following questions:  (a) If at a time, any player can play from only one team, and any number of matches can be played by any team or player, then update the above ER diagram by including cardinality and participation of each entity. (b) After alteration, highlight any errors still existing in the current ER model and correct it. (c) Convert the corrected ER model into a relational model. (d) If any derived attribute exists in the above ER model, then how would you suggest obtaining its value. If no such attribute is present, then mention that no derived attribute exists.	1 +1 +2 +1	C2	L6
Q2(b)	What is the difference between the concept of specialization and generalization? Provide example for each one with representation by using relevant symbols.	5	C2	L4

Q3(a)

Consider the two tables T1 and T2.

Table T1

P	Q	R
10	a	5
15	b	8
25	a	6

Table T2

A	B	C
10	b	6
25	c	3
10	b	5

Show the results of the following operations:

a. $T1 \bowtie_{(T1.P = T2.A)} T2$
b. $T1 \bowtie_{(T1.Q = T2.B)} T2$
c. $T1 \bowtie_{(T1.P = T2.A)} T2$
d. $T1 \bowtie_{(T1.Q = T2.B)} T2$
e. $T1 \cup T2$
f. $T1 \bowtie_{(T1.P = T2.A \text{ AND } T1.R = T2.C)} T2$

5 | **C3** | **L3**

Q3(b)

Explain each of the following key:

A. Primary Key
B. Candidate Key
C. Foreign Key
D. Super Key

Find out Primary Key, Candidate Key and Super Key from **Student** Relation. Also, explain foreign key and referential integrity constraints between **Student** Relation and **Event** Relation.

Student

Student#	Name	DOB	Registration Fee	Department
1002	Ankit	31-01-2020	39302	EIED
1035	Samiksha	05-07-1999	36493	ECED
1014	Abhinav	13-11-1997	36493	ECED
1008	Madhuri	31-01-2020	NULL	EIED
1022	Ankit	02-02-2002	32443	CSED

Event

EName	Participant
Sports2024	1002
Sports2024	1014
Sports2024	1008
Cultural Fest	1014
Cultural Fest	1002

5 | **C2** | **L1, L3**

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